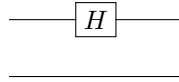

Quantum Computation and Communications (EE798V): Problem Set 3

Department of Electrical Engineering, IIT Kanpur

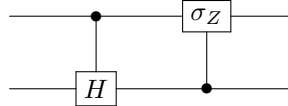
Q. 1: Consider the quantum circuit shown below.



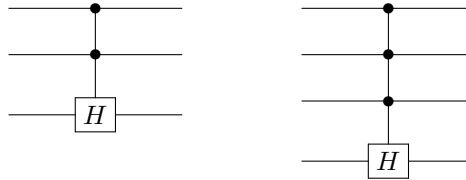
- (a) Describe how the above circuit transforms an arbitrary basis state $|jk\rangle$, $j, k \in \{0, 1\}$.
- (b) Write down the unitary matrix describing the transformation induced by the circuit. Show that the obtained matrix can be written as $H \otimes I$.
- (c) Repeat exercises (a) and (b) for the circuits shown below.



- (d) Repeat exercises (a) and (b) for the circuit shown below.



- (e) How are the matrices computed in (c) and (d) are connected?
- (f) Assume that a set of n qubits first pass through a gate represented by the unitary matrix U and then through another gate represented by the unitary matrix V . Taking hints from exercise (e), obtain a matrix (in terms of U and V) that represents the combined effect of the two gates. Using the unitary properties of U and V , show that the obtained matrix is also unitary.
- (g) Taking hints from exercise (c), obtain unitary matrices representing transformations induced by the following circuits.



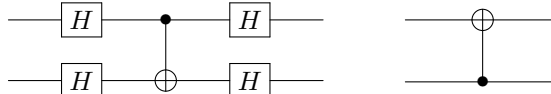
- (h) Consider a controlled- U operation (where U is a unitary matrix) that is a generalization of the circuit shown in exercise (g). The circuit acts on $(k + 1)$ qubits. The first k qubits are control qubits and the $(k + 1)$ th qubit is the target qubit. Write the matrix representation of the controlled- U operation. Prove that the obtained matrix is unitary.

Q. 2: Two quantum circuits are said to be equivalent if they induce the same basis transformation or equivalently, are represented by the same unitary matrices.

- (a) Show that the following two circuits are equivalent.



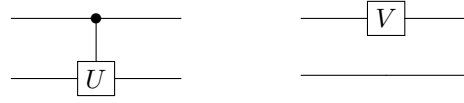
- (b) Show that the following two circuits are equivalent.



(c) Consider the following unitary matrices.

$$U = \begin{bmatrix} \exp(i\alpha) & 0 \\ 0 & \exp(i\alpha) \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 0 \\ 0 & \exp(i\alpha) \end{bmatrix}$$

Show that the following two circuits are equivalent.



Q. 3: Prove the following single qubit circuit identities.

(a) $\sigma_X \sigma_Y \sigma_X = -\sigma_Y$

(b) $H \sigma_X H = \sigma_Z$, $H \sigma_Y H = -\sigma_Y$, $H \sigma_Z H = \sigma_X$

(c) Let (n_X, n_Y, n_Z) be a real three dimensional vector of unit norm. Show that

$$(n_X \sigma_X + n_Y \sigma_Y + n_Z \sigma_Z)^2 = I$$

(d) Recall that the rotation around the Y -axis (in the Bloch sphere) by an angle θ is described by the following matrix.

$$R_Y(\theta) = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) \sigma_Y$$

Use exercises (a) and (b) to prove the following identities.

$$\sigma_X R_Y(\theta) \sigma_X = R_Y(-\theta); \quad H R_Y(\theta) H = R_Y(-\theta)$$

Q. 4: Define the following matrix.

$$R_Z(\phi) = \cos\left(\frac{\phi}{2}\right) I - i \sin\left(\frac{\phi}{2}\right) \sigma_Z$$

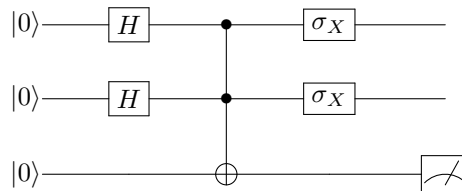
Assume that an arbitrary state $|\psi_0\rangle$ is represented in the Bloch sphere by the angles (ϕ_0, θ_0) .

(a) Prove that $R_Z(\phi)$ is unitary.

(b) Let $|\psi_1\rangle = R_Z(\phi)|\psi_0\rangle$. Show that $|\psi_1\rangle$ can be represented in the Bloch sphere by angles $(\phi_0 + \phi, \theta_0)$.

(c) Argue that $R_Z(\phi)$ induces a rotation around the Z axis (in the Bloch sphere) by an angle ϕ .

Q. 5: Describe the possible joint states that are obtainable as an outcome of the following circuit and their corresponding probabilities.



Repeat the same exercise for the following circuit.

